EDGE INTELLIGENCE LABORATORY

ASSIGNMENT 8

SUBMITTED BY

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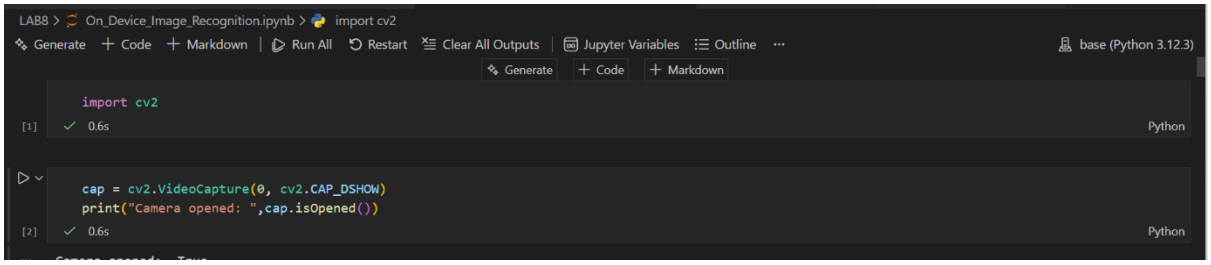
25MML0042

On Device Image Recognition

1.Importing Libraries and Checking live video from laptop camera



2.Capturing video from Laptop camera

```python
import cv2
#Open default camera(0 = laptop webcam)
cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read() #read frame from camera

    if not ret:
        print("Failed to grab frame")
        break

    cv2.imshow("laptop camera",frame) # show the frame

    #press 'q' to exit
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
```

Run by Line (F10)

19.4s

3.Loading sample video from local device

```python
import cv2

cap = cv2.VideoCapture("sample_video.mp4")

while cap.isOpened():
    ret, frame = cap.read()

    if not ret:
        break

    cv2.imshow("Video",frame)

    if cv2.waitKey(25) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

7.4s

4.Analyzing the sample offline video's Grayscale ,RGB,Negative,

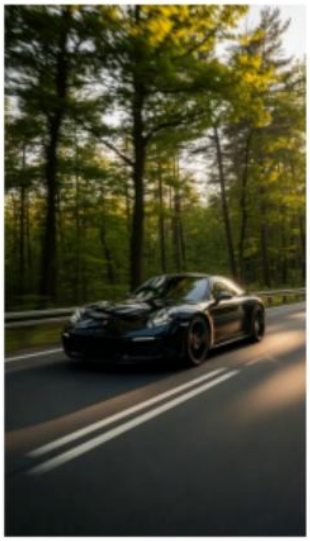
```python
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load video
cap = cv2.VideoCapture("sample_video.mp4")   # change to your video path

ret, frame = cap.read()

if not ret:
    print("Error reading video")
    cap.release()
    exit()

# Convert formats
gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
negative = 255 - frame

# Display images in output cell
plt.figure(figsize=(10,8))

plt.subplot(2,2,1)
plt.title("Original")
plt.imshow(rgb)
```

```python
plt.subplot(2,2,2)
plt.title("Gray")
plt.imshow(gray, cmap="gray")
plt.axis("off")

plt.subplot(2,2,3)
plt.title("RGB")
plt.imshow(rgb)
plt.axis("off")

plt.subplot(2,2,4)
plt.title("Negative")
plt.imshow(cv2.cvtColor(negative, cv2.COLOR_BGR2RGB))
plt.axis("off")

plt.show()

# Print pixel values of first frame
h, w, c = frame.shape

print("Frame Shape:", frame.shape)
print("\nPixel values of first frame:\n")

for i in range(h):
    for j in range(w):
        print(f"Pixel ({i},{j}) BGR:", frame[i,j])
```

✓ 25.2s

Output:

Original



Gray

RGB

Negative

Showing pixel values:

```
Pixel (1072,28) BGR: [18 20 13]
Pixel (1072,29) BGR: [19 21 14]
Pixel (1072,30) BGR: [19 21 14]
Pixel (1072,31) BGR: [18 20 13]
Pixel (1072,32) BGR: [19 18 12]
Pixel (1072,33) BGR: [19 18 12]
Pixel (1072,34) BGR: [19 18 12]
Pixel (1072,35) BGR: [19 18 12]
Pixel (1072,36) BGR: [20 19 13]
Pixel (1072,37) BGR: [20 19 13]
Pixel (1072,38) BGR: [20 19 13]
Pixel (1072,39) BGR: [20 19 13]
Pixel (1072,40) BGR: [19 18 12]
Pixel (1072,41) BGR: [19 18 12]
Pixel (1072,42) BGR: [19 18 12]
Pixel (1072,43) BGR: [19 18 12]
Pixel (1072,44) BGR: [20 19 13]
Pixel (1072,45) BGR: [20 19 13]
```

5.Loading Offline sample video and printing total frames of a video,fps,width and height of the frame.

```python
#Display Video Properties

import cv2

cap = cv2.VideoCapture("sample_video.mp4")

total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
fps = cap.get(cv2.CAP_PROP_FPS)
width = cap.get(cv2.CAP_PROP_FRAME_WIDTH)
height = cap.get(cv2.CAP_PROP_FRAME_HEIGHT)

print("Total frames:",total_frames)
print("fps:",fps)
print("width",width)
print("height:",height)

cap.release()
```

✓ 0.0s

```
Total frames: 241
fps: 24.0
width 612.0
height: 1080.0
```

6.Loading video and converting into Grayscale video:

```python
import cv2

cap = cv2.VideoCapture("sample_video.mp4")

while True:
    ret, frame = cap.read()
    if not ret:
        break

    gray = cv2.cvtColor(frame,cv2.COLOR_BGR2GRAY)
    cv2.imshow("Grayscale video:",gray)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

✓ 2.9s

7.Creating Rectangle and circle using opencv

```python
import cv2

cap = cv2.VideoCapture(0)

# Create fullscreen window
cv2.namedWindow("Shapes", cv2.WND_PROP_FULLSCREEN)
cv2.setWindowProperty("Shapes", cv2.WND_PROP_FULLSCREEN, cv2.WINDOW_FULLSCREEN)

while True:
    ret, frame = cap.read()
    if not ret:
        break

    # Draw rectangle
    cv2.rectangle(frame, (100,100),(300,300),(0,255,0),3)

    # Draw circle
    cv2.circle(frame,(200,200),50,(255,0,0),3)

    # Put text
    cv2.putText(frame,"OpenCV practice",(50,50),
                cv2.FONT_HERSHEY_SIMPLEX,1,(0,0,255),2)

    cv2.imshow("Shapes",frame)
```

```
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```
✓ 15.3s

Output:

8.Converting live video to find edge using canny operator:

Mathematical Equation:

Step 1: Convert Image to Grayscale

$$Gray = 0.299R + 0.587G + 0.114B$$

Step 2: Noise Reduction (Gaussian Blur)

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

The image is convolved with the Gaussian kernel:

$$I_s(x,y) = I(x,y) * G(x,y)$$

Step 3: Compute Image Gradient it finds $G_x and G_y$

Step 4: Gradient Magnitude (Edge Strength)

$$G = |G_x| + |G_y|$$

Step 5: Gradient Direction (Edge Orientation)

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$G(x,y) > T_H$$

$$T_L < G(x,y) < T_H$$

$$G(x,y) < T_L$$

Step 6: Non-Maximum Suppression

Steps of **Canny Edge Detection** :

1. **Convert the image to grayscale** – reduce the image from RGB to a single intensity channel.
2. **Apply Gaussian blur** – smooth the image to remove noise.
3. **Calculate image gradients** – find how quickly the intensity changes in horizontal and vertical directions.

4. **Compute gradient magnitude and direction** – determine the strength and orientation of edges.
5. **Apply non-maximum suppression** – thin the edges by keeping only the strongest pixels.
6. **Apply double thresholding** – classify pixels as strong edges, weak edges, or non-edges.
7. **Perform edge tracking by hysteresis** – keep weak edges that are connected to strong edges and remove the rest.

```python
import cv2

# Open webcam
cap = cv2.VideoCapture(0)

if not cap.isOpened():
    print("Cannot open camera")
    exit()

# Create fullscreen windows
cv2.namedWindow("Original Video", cv2.WND_PROP_FULLSCREEN)
cv2.setWindowProperty("Original Video", cv2.WND_PROP_FULLSCREEN, cv2.WINDOW_FULLSCREEN)

cv2.namedWindow("Canny Edge Detection", cv2.WND_PROP_FULLSCREEN)
cv2.setWindowProperty("Canny Edge Detection", cv2.WND_PROP_FULLSCREEN, cv2.WINDOW_FULLSCREEN)

while True:
    # Capture frame
    ret, frame = cap.read()

    if not ret:
        print("Can't receive frame")
        break
```

```
    if not ret:
        print("Can't receive frame")
        break

    # Convert to grayscale
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Gaussian blur
    blur = cv2.GaussianBlur(gray, (5,5), 0)

    # Canny edge detection
    edges = cv2.Canny(blur, 100, 200)

    # Display frames
    cv2.imshow("Original Video", frame)
    cv2.imshow("Canny Edge Detection", edges)

    # Press q to exit
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

✓ 11.3s

Output:

9.Converting Live video to Negative video:

```python
import cv2

# Open webcam
cap = cv2.VideoCapture(0)

if not cap.isOpened():
    print("Cannot open camera")
    exit()

# Create fullscreen window
cv2.namedWindow("Negative Video", cv2.WND_PROP_FULLSCREEN)
cv2.setWindowProperty("Negative Video", cv2.WND_PROP_FULLSCREEN, cv2.WINDOW_FULLSCREEN)

while True:
    ret, frame = cap.read()

    if not ret:
        print("Can't receive frame")
        break

    # Create negative image
    negative = 255 - frame
```

```python
    # Show negative video
    cv2.imshow("Negative Video", negative)

    # Press 'q' to quit
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

✓ 23.6s

Output:

Mathematical Equation

For an 8-bit image:

$$s = 255 - r$$

Where:

- r= original pixel value
- s= negative pixel value
- 255 = maximum intensity value in an 8-bit image

10.Detecting the motion if motion found edge will be shown otherwise black screen will be shown.

Mathematical Equation:

1. Grayscale Conversion

$$Gray(x, y) = 0.299R(x, y) + 0.587G(x, y) + 0.114B(x, y)$$

2. Frame Difference (Motion Detection):

$$D(x, y) = |G_t(x, y) - G_{t-1}(x, y)|$$

```python
import cv2
import numpy as np

# Open webcam
cap = cv2.VideoCapture(0)

# Fullscreen window
cv2.namedWindow("Motion Edge", cv2.WND_PROP_FULLSCREEN)
cv2.setWindowProperty("Motion Edge", cv2.WND_PROP_FULLSCREEN, cv2.WINDOW_FULLSCREEN)

# Read first frame
ret, prev_frame = cap.read()
prev_gray = cv2.cvtColor(prev_frame, cv2.COLOR_BGR2GRAY)

while True:

    ret, frame = cap.read()
    if not ret:
        break

    # Convert current frame to gray
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
```

```
    # Frame difference
    diff = cv2.absdiff(prev_gray, gray)

    # Threshold to detect motion
    _, thresh = cv2.threshold(diff, 25, 255, cv2.THRESH_BINARY)

    motion_pixels = cv2.countNonZero(thresh)

    # If motion detected
    if motion_pixels > 5000:

        # Apply Canny Edge
        edges = cv2.Canny(gray, 100, 200)

        cv2.imshow("Motion Edge", edges)

    else:
        # Show black screen
        black = np.zeros_like(gray)
        cv2.imshow("Motion Edge", black)

    # Update previous frame
    prev_gray = gray

    # Quit key
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
```

```
cap.release()
cv2.destroyAllWindows()
✓ 1m 19.0s
```

Output:

[Drive Output](#)